ManyDaughters project – Pre-analysis plan

Team

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In our approach, the identification of causal effects relies on the random assignment of child gender among single-child families. We can assume that the child's gender in one-child families is random. This avoids selection effects into having children in the first place, and possible selection effects when having more children (e.g., research has shown that parents prefer different genders among children and we cannot exclude a preference for a specific gender, which, loosely speaking, would result in families having children until a particular gender or gender ratio is realized). We opted for this approach due to data availability issues, which prevent us from exploring the longitudinal nature of the SOEP for identification (e.g., outcomes and treatments not always available in a panel-like setting, making panel-based approaches infeasible). We acknowledge that this approach only identifies very local effects.

All robust standard errors will be obtained using *vce(robust)*, which estimates Huber-White-Sandwich type standard errors (HC1). *vce(cluster clustvar)* will be used for clustered standard errors. No weighting will be applied.

1 Hypothesis 1

Having daughters leads to stronger support for gender equality (as measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for respondents with daughters) > (X for respondents without daughters).

- **Variables**:

  Outcome variable:

  - Additive index of gender equality attitudes (plj0621–plj0625) in 2016.
  - Recode items plj0623, plj0624, and plj0625 such that higher values indicate more support for gender equality:

    $$\text{reversed} = (\max + \min) - \text{value}$$

  Treatment variable (*OnlyChildIsDaughter*):

  - 1 if the respondent has exactly one child in 2016 and that child is female.
  - 0 if exactly one child and that child is male.
  - Based on bgp15307a (2016) / pld0026.

  Control variables:

  - Respondent's year of birth (ple0010_v2 / gebjahr)
  - Gender (sex)
  - Education (pgisced97)
  - Employment status (pgemplst)
  - Marital status (pgfamstd, dichotomized: coded as one if married (1); married, but separated (2); husband/wife abroad (6); civil union living together or apart (7 and 8); coded as zero if single(3); divorced(4); widowed(5))

- **Inclusion/exclusion criteria**

  - Only include year 2016 data.

  - Exclude:
    * Cases without children or with more than one child.
    * Cases for which any of the variable values are missing.

- **Exact analysis**

  Regression model:

$$X_i = \beta_0 + \beta_1 \,\text{OnlyChildIsDaughter}_i + \gamma' C_i + \varepsilon_i$$

  , where C_i denotes control variables, ε_i the error term, and i individual respondents.

  - OLS with robust standard errors.

- **Interpretation of results**:
  - Statistic: t-value of β_1.
  - Decision: $p < 0.05$ (two-sided).

2 Hypothesis 2

Having daughters leads to more left-leaning political preferences. That is, denoting Y as a measure of left-leaning political preferences, (Y for respondents with daughters) > (Y for respondents without daughters).

Challenge: Treatment measured in 2001, 2006, 2011, 2016 (pld0026); outcome measured in 2005, 2009, 2014, 2019.

We will consider multiple years. If there is one daughter or son in 2001, we will create a data point with treatment information from 2001 and outcome information from 2005; similarly for 2006/2009, 2011/2014, and 2016/2019. We will cluster standard errors at the respondent level to account for participants appearing multiple times in the data.

- **Variables**: Treatment (in the respective years) and control variables as in H1, plus treatment/outcome pair fixed effects (time).

  Outcome variable:

  - Political preferences (plh0004 in 2005, 2009, 2014, 2019).
  - Reverse values, such that higher values indicate more left-leaning preferences:

$$\text{reversed} = (\max + \min) - \text{value}$$

- **Inclusion/exclusion criteria**

  - Year pairs between treatment and outcome as described above (note: potential misclassification if additional children born after treatment but before outcome was measured.)
  - Same exclusion criteria as H1.

- **Exact analysis**

  Regression model:

$$Y_{it} = \beta_0 + \beta_1 \,\text{OnlyChildIsDaughter}_{it} + \gamma' C_{it} + \delta_t + \varepsilon_{it}$$

  - OLS with robust standard errors clustered at the respondent level.

- **Interpretation of results**:
  - Same as for H1.

3 Hypothesis 3

Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes). That is, denoting Z as a measure of gender equality in the household, (Z for respondents with daughters) > (Z for respondents without daughters).

- **Variables**: Treatment variable (in the respective years) as in H1, plus fixed effects for years. No control variables.

  Outcome variable: Absolute value of difference between individual cumulated hours worked in household (childcare, chores, repair work) for partners of different genders in the same household.

  $$Z_{jt} = |\text{TotalHoursMale}_{it} - \text{TotalHoursFemale}_{it}|$$

  - Sum of hours used for three household-related activities: childcare, chores, and repair work.
  - Only hours worked on weekdays because data on Saturdays and Sundays do not seem to be collected every year.
  - Variable names: pli0044_h (childcare), pli0043_h (chores), pli0049_h (repair work).
  - Yearly data: All waves where both hours and child gender are available (i.e., 2001, 2006, 2011, 2016).

- **Inclusion/exclusion criteria**

  - Only use waves where hours and child gender are available (e.g., 2001, 2006, 2011, 2016).
  - Same exclusion criteria as for H1.
  - Exclude cases for whom no partner was identified (partner identification via parid from ppathl and pgpartnr from pgen).
  - Partners must live together (based on pgfamstd, values 1 (married) and 7 (living together in civil union) counted as 1, otherwise 0)

- **Exact analysis**

  Regression model:

  $$Z_{jt} = \beta_0 + \beta_1 \,\text{OnlyChildIsDaughter}_{jt} + \delta_t + \varepsilon_{jt}$$

  , where j is a partnership dyad.

  - OLS with robust standard errors clustered at the household level.

- **Interpretation of results**:

  - Same as for H1.

4 Hypothesis 4

Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes). That is, denoting X as a measure of support for gender equality, (X for male respondents with daughters - X for male respondents without daughters) > (X for female respondents with daughters - X for female respondents without daughters).

- **Variables**:

  - Same as H1 plus interaction term:

    $$\text{OnlyChildIsDaughter} \times \text{MaleRespondent}$$

- **Inclusion/exclusion criteria**

  - Same as for H1.

- **Exact analysis**

  Regression model:

$$X_i = \beta_0 + \beta_1 \,\text{OnlyChildIsDaughter}_i + \beta_2 \,\text{MaleRespondent}_i + \beta_3 \,(\text{OnlyChildIsDaughter}_i \times \text{MaleRespondent}_i) + \gamma' C_i + \varepsilon_i$$

  - OLS with robust standard errors.

- **Interpretation of results**:

  - Same as for H1 (but here, the cofficient of interest ist β_3).